

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027

Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

TherAlign Health, Founding Software Engineer Intern | *Remote* Jul 2026 – Present

- Founding engineer building full-stack, user-facing features (Next.js, Firebase/GCP) on a provider-facing **SMART on FHIR** clinical decision support app that surfaces covered, guideline-backed medication alternatives in the EHR prescribing flow to prevent prior authorization.
- Extending a **Firebase Cloud Functions** backend and its APIs and data flows (**RxNorm** normalization, formulary/prior-auth logic, PubMed/guideline evidence), with **Gemini** constrained to synthesis while the backend deterministically controls eligibility; developing features end to end across the stack.

Culture & Morality Lab, UMass Amherst, Lab Assistant | *Amherst, MA* Jul 2026 – Present

- Sole engineer building the lab's CCR text-analysis platform (Python, **FastAPI**, React, sentence-transformers), productionizing a published NLP method while owning architecture, APIs, and all technical decisions (delegated by the PI).
- Built versioned data infrastructure and APIs behind it: a YAML model registry, an append-only construct library (**99 constructs from 38 questionnaires**), structured data-quality warnings, and **40 hermetic backend tests**.

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, with REST APIs and data flows processing **10,000+ monthly triage submissions** via Mayden iaptus integrations; contributed to **\$340K+** in revenue.
- Architected a full-stack, user-facing ML data annotation platform (React, Node.js, MongoDB) used daily by the AI team, cutting **100+ hours/month** of manual labeling work and lifting throughput by **60%**.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a multi-stage agentic web-discovery service (FastAPI) that turns open-ended queries into structured entity tables via typed retrieval planning, reranking, and deterministic extraction with LLM fallback and **cell-level provenance**; hardened with **150+ tests**.

cf-ai-research-scout – Edge-Native AI Assistant (GitHub) TypeScript, Cloudflare Workers AI, Durable Objects, D1

- Built an edge-native, full-stack AI assistant end to end in **TypeScript** on Cloudflare Workers: an agent that fetches web sources and streams multi-step research digests over WebSocket, with durable workflows and D1 storage that survive restarts and tool failures.

Sonare – Offline Sign / Speech App (GitHub) React, Electron, FastAPI, MediaPipe, whisper.cpp

- Built a cross-platform desktop app (React/Electron frontend, FastAPI backend) with an on-device pipeline, designing async APIs and queues to keep the UI responsive while running fully offline; shipped end to end under hackathon time pressure.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages: JavaScript, TypeScript, Python, SQL, C++, Java, Bash

Frontend: React, Next.js, Redux, Tailwind CSS, HTML/CSS, Vite

Backend & APIs: Node.js, Express, FastAPI, Fastify, REST APIs, async services

Databases & Tests: MongoDB, PostgreSQL, SQLite, Redis, pytest, integration testing

Cloud & DevOps: AWS, GCP, Azure, Docker, Git, GitHub Actions, CI/CD, Cloudflare Workers

Open Source & Community

- Built **Underdogs Fitness** (Website), a production MERN platform with Stripe payments; live **3+ years**. - Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.